

Statistical Analysis of CICIDS2017 Network Intrusion Data

Multiple Linear Regression & Logistic Regression

Statistics II — IDC 5205

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Data Description and Research Questions

The CICIDS2017 Dataset

CICIDS2017 is a public network intrusion detection dataset from the Canadian Institute for Cybersecurity. Each row represents one network flow (a connection between two computers), with 79 columns describing packet sizes, timing, TCP flags, and routing. Some flows are normal benign traffic; others are attacks (DoS, port scans, brute-force) that researchers ran on purpose and labeled accordingly.

Source: <https://www.unb.ca/cic/datasets/ids-2017.html>

Research Questions

1. **MLR Question:** Among benign traffic only, which network flow features best predict the total byte volume (Flow.Bytes)? A multiple linear regression quantifies the relationship between packet-level features and flow size, building a statistical baseline of “normal” traffic shape.
2. **Logistic Question:** Which features distinguish attack flows from benign flows? A logistic regression model identifies the key risk factors and classifies traffic with measurable accuracy (ROC/AUC, sensitivity, specificity).

Data Preparation

We sampled ~250 rows per raw CSV file (8 files, each from a different day), cleaned column names, dropped infinities and NAs, and used **Stepwise BIC** for variable selection on two separate feature sets:

- **MLR features** — Stepwise BIC with `Flow.Bytes` as the response on benign-only data → 8 predictors selected.
- **Logistic features** — Stepwise BIC with `Attack_Flag` as the response on all data → 13 predictors selected (2 later dropped for high VIF).

Highly correlated pairs ($|r| > 0.9$) were pruned before selection to ensure model stability.

```
library(readr)
library(dplyr)
library(ggplot2)

# Load pre-exported datasets
mlr_data <- read_csv("mlr_data.csv", show_col_types = FALSE)
glm_data <- read_csv("glm_data.csv", show_col_types = FALSE)

cat(sprintf("MLR dataset (benign): %d rows, %d predictors + Flow.Bytes\n",
            nrow(mlr_data), ncol(mlr_data) - 1))

## MLR dataset (benign): 1611 rows, 8 predictors + Flow.Bytes

cat(sprintf("GLM dataset (all): %d rows, %d predictors + Attack_Flag\n",
            nrow(glm_data), ncol(glm_data) - 1))

## GLM dataset (all): 1997 rows, 13 predictors + Attack_Flag

cat(sprintf("Attack rate: %.1f%%\n", 100 * mean(glm_data$Attack_Flag)))

## Attack rate: 19.3%
```

Descriptive Analysis

Summary Statistics

```
cat("--- MLR Dataset Summary ---\n")
```

```
## --- MLR Dataset Summary ---
```

```
print(summary(mlr_data))
```

```
##      Flow.Bytes      act_data_pkt_fwd      PSH.Flag.Count      Avg.Bwd.Segment.Size
##  Min.      :      0      Min.      :  0.000      Min.      :0.0000      Min.      :   0.0
## 1st Qu.:     16      1st Qu.:  0.000      1st Qu.:0.0000      1st Qu.:   6.0
## Median :    224      Median :  1.000      Median :0.0000      Median :  77.0
## Mean   :   11091      Mean   :  4.471      Mean   :0.2508      Mean   : 154.6
## 3rd Qu.:    497      3rd Qu.:  2.000      3rd Qu.:1.0000      3rd Qu.: 141.3
## Max.    :10716095      Max.    :2626.000      Max.    :1.0000      Max.    :2269.5
## Bwd.Packet.Length.Std Bwd.Packet.Length.Min min_seg_size_forward
##  Min.      :  0.0      Min.      :  0.00      Min.      :20.00
## 1st Qu.:   0.0      1st Qu.:   0.00      1st Qu.:20.00
## Median :   0.0      Median :   6.00      Median :20.00
## Mean   : 116.2      Mean   :  53.01      Mean   :25.45
## 3rd Qu.:   0.0      3rd Qu.:  98.00      3rd Qu.:32.00
## Max.    :2218.0      Max.    :1442.00      Max.    :44.00
##      Active.Std      Active.Max
##  Min.      :      0      Min.      :      0
## 1st Qu.:      0      1st Qu.:      0
## Median :      0      Median :      0
## Mean   :   27768      Mean   :   122655
## 3rd Qu.:      0      3rd Qu.:      0
## Max.    :3647820      Max.    :13700000
```

```
cat("\n--- GLM Dataset Summary ---\n")
```

```
##
```

```
## --- GLM Dataset Summary ---
```

```
print(summary(glm_data))
```

```
##      Attack_Flag      Fwd.Packet.Length.Min      Packet.Length.Variance
##  Min.      :0.0000      Min.      :  0.00      Min.      :0.000e+00
## 1st Qu.:0.0000      1st Qu.:   0.00      1st Qu.:0.000e+00
## Median :0.0000      Median :   6.00      Median :5.808e+02
## Mean   :0.1933      Mean   :  19.83      Mean   :5.499e+05
## 3rd Qu.:0.0000      3rd Qu.:  37.00      3rd Qu.:1.966e+04
## Max.    :1.0000      Max.    :1375.00      Max.    :1.680e+07
## Avg.Fwd.Segment.Size URG.Flag.Count      Active.Std      Down.Up.Ratio
##  Min.      :  0.00      Min.      :0.0000      Min.      :      0      Min.      :0.0000
## 1st Qu.:   6.00      1st Qu.:0.0000      1st Qu.:      0      1st Qu.:0.0000
## Median :  32.00      Median :0.0000      Median :      0      Median :1.0000
## Mean   :  57.73      Mean   :0.1052      Mean   :  22400      Mean   :0.7176
## 3rd Qu.:  48.22      3rd Qu.:0.0000      3rd Qu.:      0      3rd Qu.:1.0000
## Max.    :2595.20      Max.    :1.0000      Max.    :3647820      Max.    :6.0000
## Destination.Port      Idle.Min      Avg.Bwd.Segment.Size      Bwd.IAT.Total
##  Min.      :      3      Min.      :      0      Min.      :  0.0      Min.      :      0
## 1st Qu.:   53      1st Qu.:      0      1st Qu.:   6.0      1st Qu.:      0
```

```
## Median : 80      Median : 0      Median : 68.0      Median : 3
## Mean : 7808     Mean : 5546107 Mean : 289.8     Mean : 9449631
## 3rd Qu.: 443    3rd Qu.: 0      3rd Qu.: 161.0   3rd Qu.: 84116
## Max. :64972     Max. :119594809 Max. :3808.4     Max. :120000000
## Init_Win_bytes_backward PSH.Flag.Count Bwd.IAT.Max
## Min. : -1      Min. :0.000 Min. : 0
## 1st Qu.: -1     1st Qu.:0.000 1st Qu.: 0
## Median : -1     Median :0.000 Median : 3
## Mean : 1674     Mean :0.315 Mean : 3793101
## 3rd Qu.: 229    3rd Qu.:1.000 3rd Qu.: 51691
## Max. :65535     Max. :1.000 Max. :119626102
```

Density Plots: Attack vs Benign

Overlay density plots show attack flows are more spread and heavy-tailed across key features.

```
glm_data$Label <- ifelse(glm_data$Attack_Flag == 1, "Attack", "Benign")
features_to_plot <- c("Destination.Port", "Avg.Fwd.Segment.Size",
                     "Avg.Bwd.Segment.Size", "Packet.Length.Variance")

par(mfrow = c(2, 2))
for (feat in features_to_plot) {
  benign_vals <- glm_data[[feat]][glm_data$Attack_Flag == 0]
  attack_vals <- glm_data[[feat]][glm_data$Attack_Flag == 1]
  dens_b <- density(benign_vals, na.rm = TRUE)
  dens_a <- density(attack_vals, na.rm = TRUE)
  ymax <- max(c(dens_b$y, dens_a$y))
  xrange <- range(c(dens_b$x, dens_a$x))
  plot(dens_b, main = paste("Density of", feat), xlab = feat,
       col = "steelblue", lwd = 2, xlim = xrange, ylim = c(0, ymax))
  lines(dens_a, col = "firebrick", lwd = 2)
  legend("topright", legend = c("Benign", "Attack"),
       col = c("steelblue", "firebrick"), lwd = 2, bty = "n")
}
```

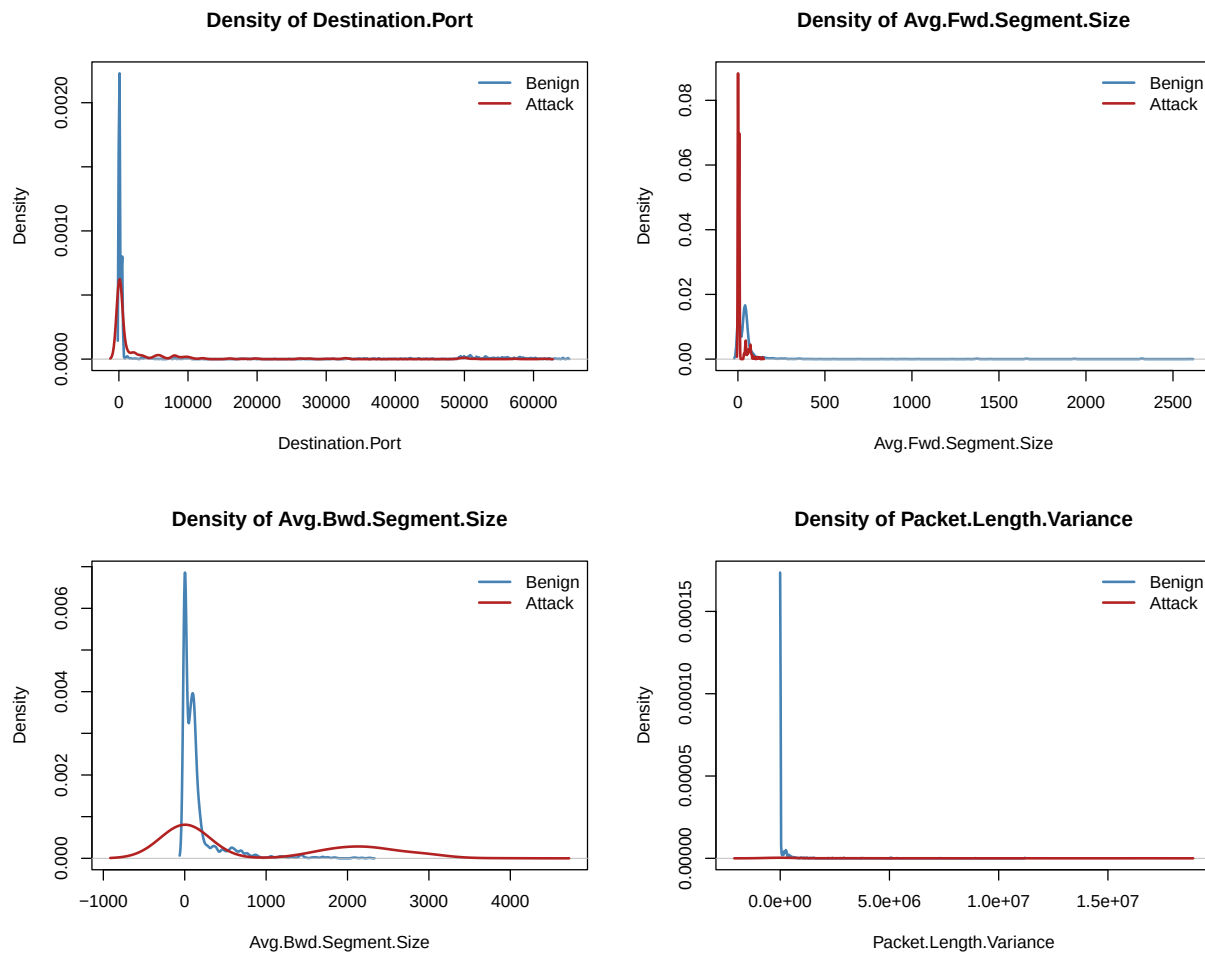


Figure 1: Attack vs Benign Density Overlays

```
par(mfrow = c(1, 1))
```

Correlation Heatmaps

```
par(mfrow = c(1, 2))
cor_mlr <- cor(mlr_data)
heatmap(cor_mlr, symm = TRUE, main = "MLR Predictor Correlations\n(Benign Traffic)",
        col = colorRampPalette(c("navy", "white", "firebrick"))(100))
```

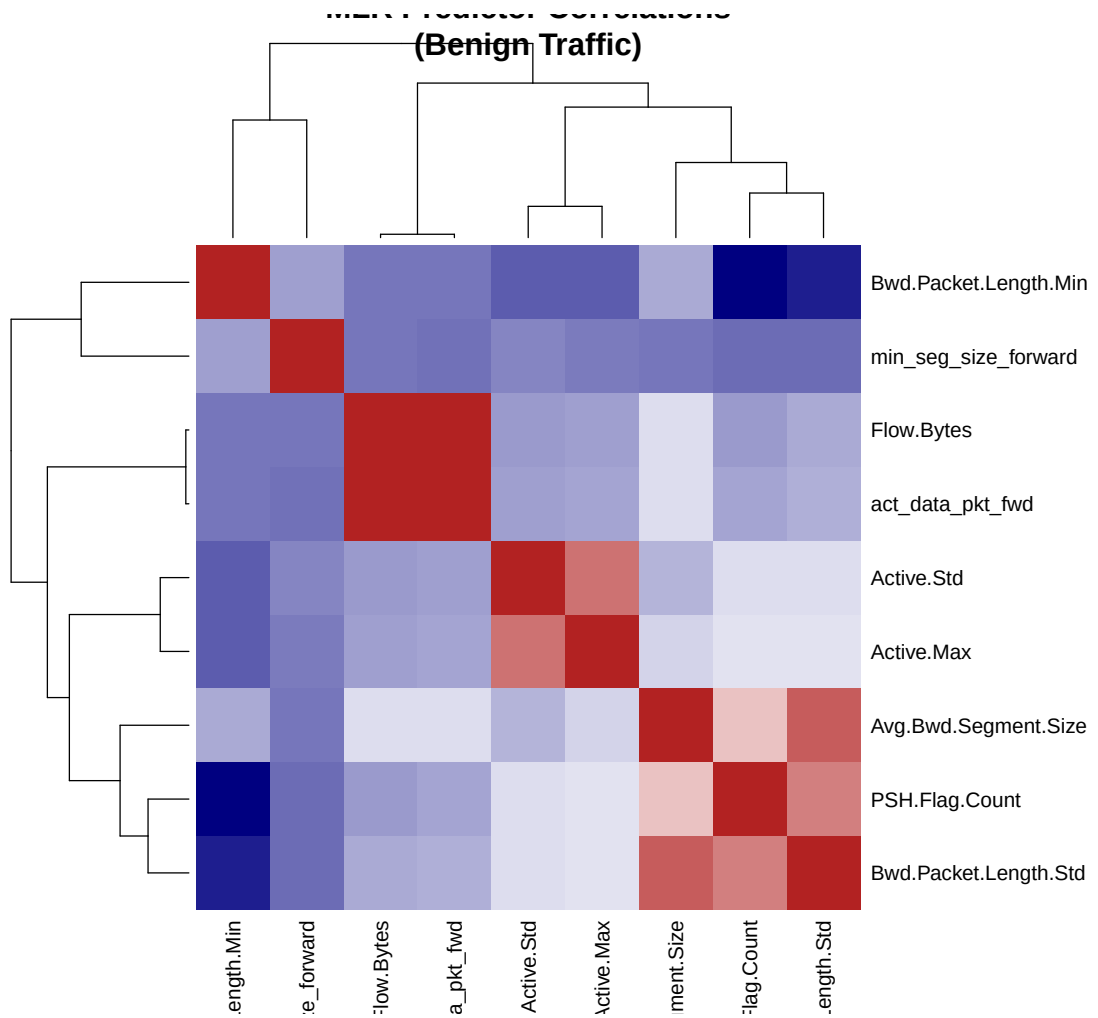


Figure 2: Correlation Heatmaps

```
cor_glm <- cor(dplyr::select(glm_data, -Attack_Flag, -Label))
heatmap(cor_glm, symm = TRUE, main = "GLM Predictor Correlations\n(All Traffic)",
        col = colorRampPalette(c("navy", "white", "firebrick"))(100))
```

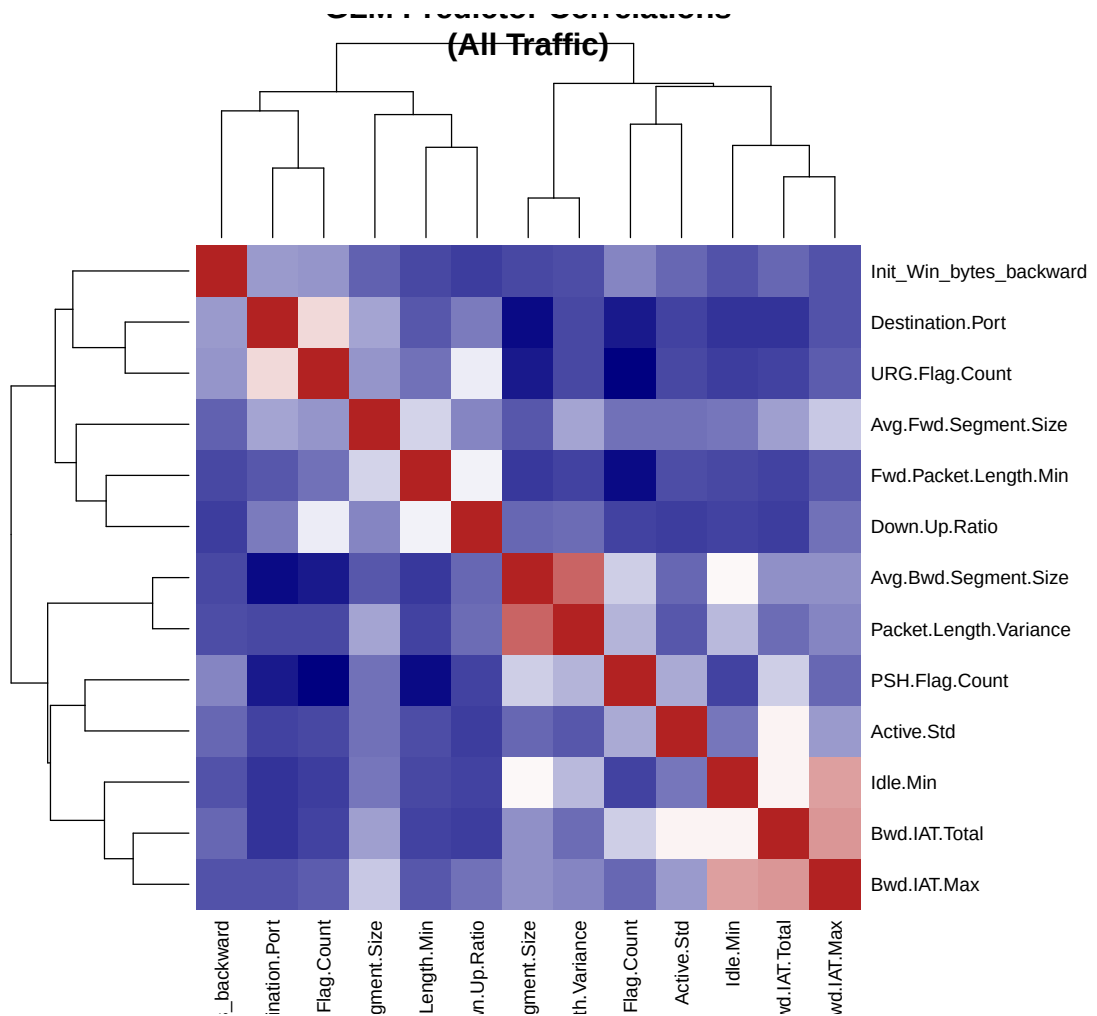


Figure 3: Correlation Heatmaps

```
par(mfrow = c(1, 1))
```

Two Categorical Variables: Port Category vs Attack Flag

IANA port ranges: well-known (0–1023), registered (1024–49151), dynamic (49152–65535). A chi-square test confirms the relationship between destination port category and attack status.

```
glm_data$Port_Category <- cut(
  glm_data$Destination.Port,
  breaks = c(-Inf, 1023, 49151, Inf),
  labels = c("Well-known", "Registered", "Dynamic"))

port_attack_tab <- table(Port_Category = glm_data$Port_Category,
  Attack = ifelse(glm_data$Attack_Flag == 1, "Yes", "No"))

print(port_attack_tab)
```

```
##           Attack
## Port_Category  No  Yes
## Well-known    1268 266
## Registered     139 111
## Dynamic        204  9
```

```
cat("\nRow-wise attack rate (%):\n")
```

```
##
## Row-wise attack rate (%):
```

```
print(round(prop.table(port_attack_tab, 1) * 100, 1))
```

```
##           Attack
## Port_Category  No  Yes
## Well-known    82.7 17.3
## Registered    55.6 44.4
## Dynamic       95.8  4.2
```

```
chi <- chisq.test(port_attack_tab)
cat(sprintf("\nChi-square test: X^2 = %.2f, df = %d, p = %.3e\n",
  chi$statistic, chi$parameter, chi$p.value))
```

```
##
## Chi-square test: X^2 = 135.83, df = 2, p = 3.201e-30
```

Comparative Boxplots

```
top_features <- c("Avg.Fwd.Segment.Size", "Avg.Bwd.Segment.Size",
  "Packet.Length.Variance", "Destination.Port",
  "Down.Up.Ratio", "Init_Win_bytes_backward")

par(mfrow = c(2, 3))
for (feat in top_features) {
  boxplot(glm_data[[feat]] ~ glm_data$Attack_Flag,
    names = c("Benign", "Attack"), main = feat,
    ylab = feat, xlab = "",
    col = c("steelblue", "firebrick"))
}
```

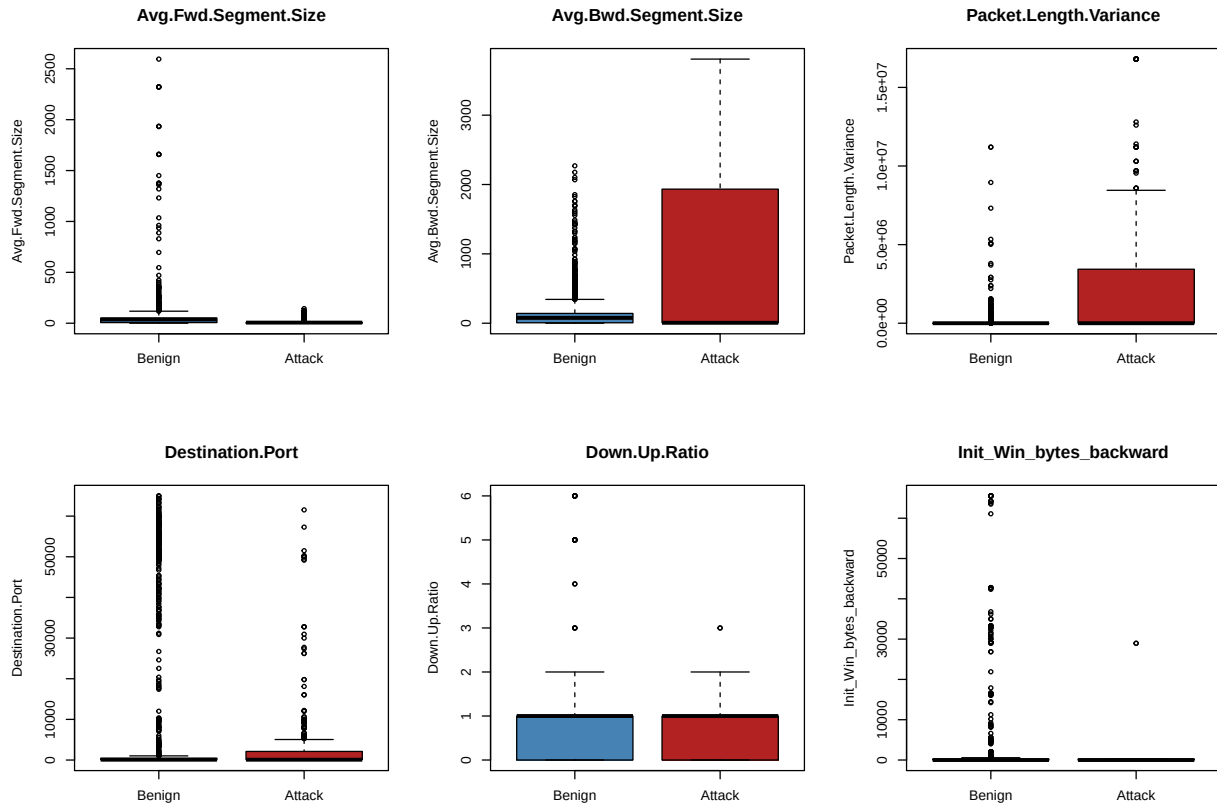



Figure 4: Benign vs Attack Boxplots

```
par(mfrow = c(1, 1))

group_stats <- glm_data |>
  dplyr::group_by(Attack_Flag) |>
  dplyr::summarise(dplyr::across(dplyr::all_of(top_features),
    list(mean = ~mean(.x, na.rm = TRUE),
          sd = ~sd(.x, na.rm = TRUE)))) |>
  as.data.frame()
print(group_stats)
```

##	Attack_Flag	Avg.Fwd.Segment.Size_mean	Avg.Fwd.Segment.Size_sd
## 1	0	68.48325	216.06820
## 2	1	12.82526	23.47649

##	Avg.Bwd.Segment.Size_mean	Avg.Bwd.Segment.Size_sd	Packet.Length.Variance_mean
## 1	154.5855	277.5848	119027.6
## 2	854.0098	1114.6153	2348052.4

##	Packet.Length.Variance_sd	Destination.Port_mean	Destination.Port_sd
## 1	613693.6	8846.313	19146.475
## 2	4020224.3	3476.953	9250.279

##	Down.Up.Ratio_mean	Down.Up.Ratio_sd	Init_Win_bytes_backward_mean
## 1	0.7206704	0.7419555	2016.7529
## 2	0.7046632	0.5728299	244.6477

##	Init_Win_bytes_backward_sd
## 1	2016.7529
## 2	244.6477

## 1	8854.501
## 2	2078.172

Multiple Linear Regression

Outlier Detection and Pruning

Before fitting the final model, we checked for high-leverage observations using hat values. The initial model (fit on all 1611 benign flows) revealed one extreme point:

```
library(car)

# Fit initial model to detect outliers
mlr_init <- lm(Flow.Bytes ~ ., data = mlr_data)
hatv <- hatvalues(mlr_init)
p <- length(coef(mlr_init))
n <- nrow(mlr_data)
threshold <- 2 * p / n
outlier_row <- which.max(hatv)

cat("--- Initial Model Leverage Diagnostics ---\n")

## --- Initial Model Leverage Diagnostics ---
cat(sprintf("Leverage threshold (2p/n): %.4f\n", threshold))

## Leverage threshold (2p/n): 0.0112
cat(sprintf("Worst observation - row %d:\n", outlier_row))

## Worst observation - row 1198:
cat(sprintf("  Leverage = %.4f (%.1f × threshold)\n",
            hatv[outlier_row], hatv[outlier_row] / threshold))

##   Leverage = 0.9901 (88.6 × threshold)
cat(sprintf("  Flow.Bytes = %d (next largest: %d)\n",
            mlr_data$Flow.Bytes[outlier_row],
            max(mlr_data$Flow.Bytes[-outlier_row])))

##   Flow.Bytes = 10716095 (next largest: 977134)
cat(sprintf("  act_data_pkt_fwd = %d (median = %d, mean = %.1f)\n",
            mlr_data$act_data_pkt_fwd[outlier_row],
            median(mlr_data$act_data_pkt_fwd), mean(mlr_data$act_data_pkt_fwd)))

##   act_data_pkt_fwd = 2626 (median = 1, mean = 4.5)
cat("\nThe initial R2 was dominated by this single point.\n")

##
## The initial R2 was dominated by this single point.
cat(sprintf("  With outlier:   R2 = %.4f\n", summary(mlr_init)$r.squared))

##   With outlier:   R2 = 0.9911
```

This one flow — a 10.7 million byte transfer with 2626 active data packets — has leverage 88 times the threshold. It is a genuine observation (large file transfers exist on real networks), but it is so far from the rest of the data that a single linear model cannot simultaneously describe it and the other 1610 flows. We remove it and refit on the remaining 1610 benign flows.

```
mlr_clean <- mlr_data[-outlier_row, ]
cat(sprintf("Clean data: %d rows\n", nrow(mlr_clean)))
```

```
## Clean data: 1610 rows
```

Model: Flow.Bytes ~ . using the 8 BIC-selected predictors on benign traffic ($n = 1610$ after pruning).

```
mlr_model <- lm(Flow.Bytes ~ ., data = mlr_clean)
s_mlr <- summary(mlr_model)
print(s_mlr)
```

```
##
## Call:
## lm(formula = Flow.Bytes ~ ., data = mlr_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -109803   -1705     135    3148   649548
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -8.449e+03  2.323e+03  -3.637 0.000285 ***
## act_data_pkt_fwd  1.832e+03  8.173e+01  22.409 < 2e-16 ***
## PSH.Flag.Count   -6.225e+03  1.819e+03  -3.422 0.000637 ***
## Avg.Bwd.Segment.Size  1.041e+02  4.677e+00  22.251 < 2e-16 ***
## Bwd.Packet.Length.Std -7.373e+01  5.050e+00 -14.599 < 2e-16 ***
## Bwd.Packet.Length.Min -1.029e+02  7.848e+00 -13.117 < 2e-16 ***
## min_seg_size_forward  2.652e+02  8.618e+01   3.077 0.002125 **
## Active.Std         -3.849e-02  4.634e-03  -8.306 < 2e-16 ***
## Active.Max         1.302e-02  1.247e-03  10.438 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21030 on 1601 degrees of freedom
## Multiple R-squared:  0.6249, Adjusted R-squared:  0.623
## F-statistic: 333.4 on 8 and 1601 DF,  p-value: < 2.2e-16
```

ANOVA (Overall F-Test)

```
anova(mlr_model)
```

```
## Analysis of Variance Table
##
## Response: Flow.Bytes
##
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
act_data_pkt_fwd	1	8.6010e+11	8.6010e+11	1945.6112	< 2.2e-16 ***
PSH.Flag.Count	1	6.8588e+09	6.8588e+09	15.5151	8.537e-05 ***
Avg.Bwd.Segment.Size	1	1.3174e+11	1.3174e+11	298.0118	< 2.2e-16 ***
Bwd.Packet.Length.Std	1	4.6333e+10	4.6333e+10	104.8073	< 2.2e-16 ***
Bwd.Packet.Length.Min	1	8.0893e+10	8.0893e+10	182.9861	< 2.2e-16 ***
min_seg_size_forward	1	4.4178e+09	4.4178e+09	9.9933	0.001601 **
Active.Std	1	4.1504e+08	4.1504e+08	0.9389	0.332718
Active.Max	1	4.8167e+10	4.8167e+10	108.9569	< 2.2e-16 ***
Residuals	1601	7.0776e+11	4.4207e+08		

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Confidence Intervals

```
confint(mlr_model, level = 0.95)
```

```
##                2.5 %      97.5 %
## (Intercept)    -1.300650e+04 -3.892136e+03
## act_data_pkt_fwd  1.671243e+03  1.991878e+03
## PSH.Flag.Count  -9.793332e+03 -2.656970e+03
## Avg.Bwd.Segment.Size  9.490163e+01  1.132506e+02
## Bwd.Packet.Length.Std -8.363792e+01 -6.382593e+01
## Bwd.Packet.Length.Min -1.183284e+02 -8.754250e+01
## min_seg_size_forward  9.615739e+01  4.342465e+02
## Active.Std       -4.758192e-02 -2.940258e-02
## Active.Max       1.057234e-02  1.546501e-02
```

Multicollinearity (VIF)

```
vif_values <- car::vif(mlr_model)
print(vif_values)
```

```
##      act_data_pkt_fwd      PSH.Flag.Count  Avg.Bwd.Segment.Size
##      1.720699          2.261688          5.937063
## Bwd.Packet.Length.Std Bwd.Packet.Length.Min  min_seg_size_forward
##      6.830515          1.769534          1.038479
##      Active.Std          Active.Max
##      2.406375          2.545160
```

```
cat(sprintf("\nMax VIF: %.2f\n", max(vif_values)))
```

```
##
## Max VIF: 6.83
```

Model Diagnostics

```
par(mfrow = c(2, 2))
plot(mlr_model, which = 1, main = "Residuals vs Fitted\n(Linearity & Homoscedasticity)")
abline(h = 0, col = "red", lty = 2)
plot(mlr_model, which = 2, main = "Normal Q-Q\n(Normality of Residuals)")
plot(mlr_model, which = 3, main = "Scale-Location\n(Constant Variance)")
plot(mlr_model, which = 5, main = "Residuals vs Leverage\n(Influential Points)")
```

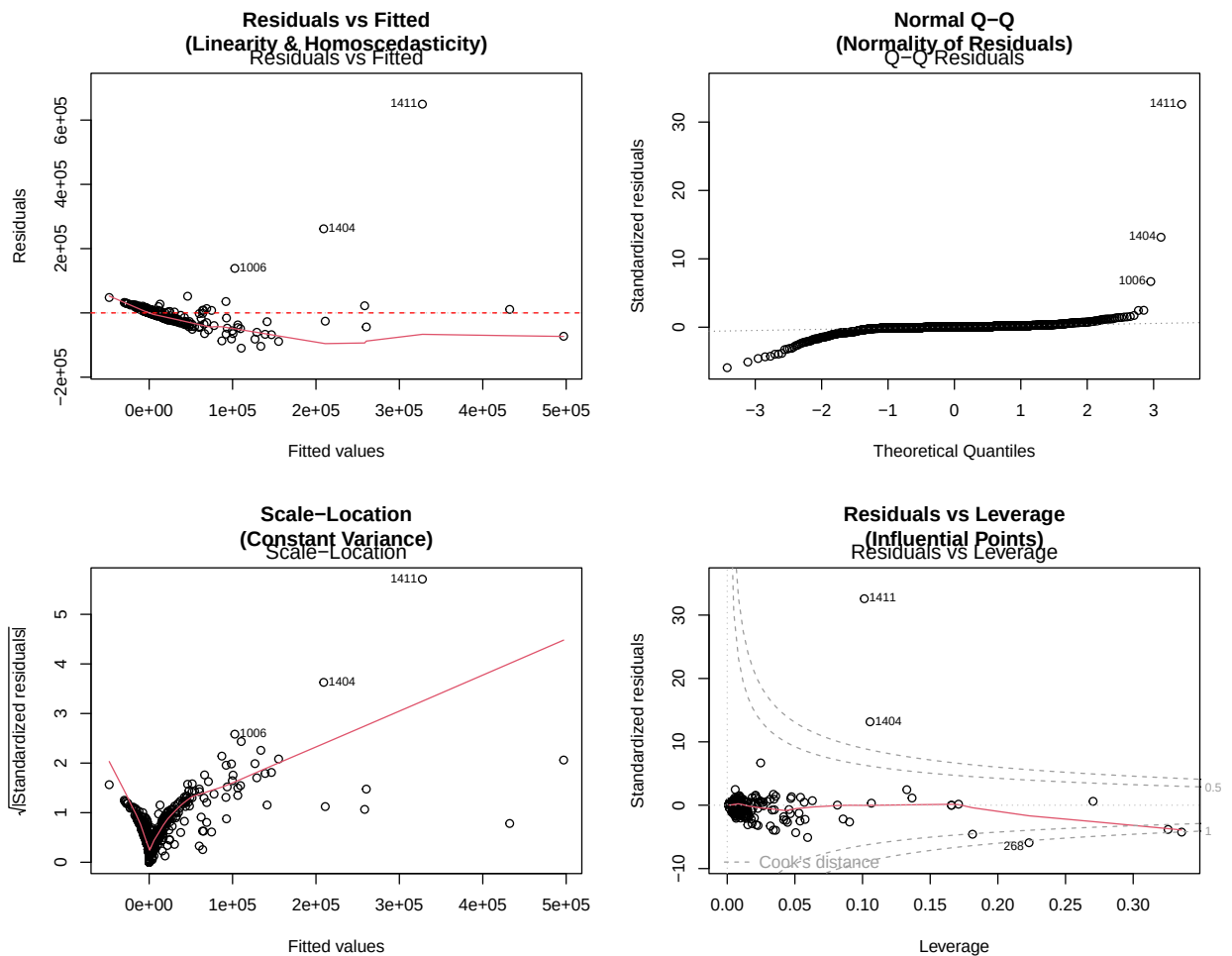


Figure 5: MLR Diagnostic Plots (Clean Data)

```
par(mfrow = c(1, 1))
```

Heteroscedasticity Test (NCV)

```
ncv_test <- ncvTest(mlr_model)
print(ncv_test)
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 61438.31, Df = 1, p = < 2.22e-16
```

Normality Test (Shapiro-Wilk)

```
shapiro_test <- shapiro.test(residuals(mlr_model))
print(shapiro_test)
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(mlr_model)
## W = 0.26274, p-value < 2.2e-16
```

Influence Diagnostics

```
cooks_d <- cooks.distance(mlr_model)
n_clean <- nrow(mlr_clean)
cat(sprintf("Cook's distance > 4/n: %d observations\n", sum(cooks_d > 4 / n_clean)))
```

```
## Cook's distance > 4/n: 57 observations
```

```
hat_values <- hatvalues(mlr_model)
p <- length(coef(mlr_model))
cat(sprintf("High leverage (> 2p/n): %d observations\n",
            sum(hat_values > 2 * p / n_clean)))
```

```
## High leverage (> 2p/n): 91 observations
```

```
dfbetas_vals <- dfbetas(mlr_model)
large_dfbetas <- which(abs(dfbetas_vals) > 2 / sqrt(n_clean), arr.ind = TRUE)
cat(sprintf("Large DFBETAS (> 2/sqrt(n)): %d instances\n", nrow(large_dfbetas)))
```

```
## Large DFBETAS (> 2/sqrt(n)): 377 instances
```

```
rstud <- rstudent(mlr_model)
cat(sprintf("|rstudent| > 3: %d observations\n", sum(abs(rstud) > 3)))
```

```
## |rstudent| > 3: 15 observations
```

Robustness Check: Log-Transformed Response

The diagnostics above flag heteroscedasticity and non-normality. We fit $\log(\text{Flow.Bytes} + 1)$ as a robustness check and compare the diagnostic plots.

```
mlr_log <- mlr_clean
mlr_log$Flow.Bytes <- log1p(mlr_log$Flow.Bytes)

mlr_model_log <- lm(Flow.Bytes ~ ., data = mlr_log)

ncv_log <- ncvTest(mlr_model_log)
shapiro_log <- shapiro.test(residuals(mlr_model_log))

cat(sprintf("Log model R-squared: %.4f (raw: %.4f)\n",
            summary(mlr_model_log)$r.squared, s_mlr$r.squared))
```

```
## Log model R-squared: 0.5686 (raw: 0.6249)
```

```
cat(sprintf("NCV test p-value -- raw: %.3e | log: %.3e\n",
            ncv_test$p, ncv_log$p))
```

```
## NCV test p-value -- raw: 0.000e+00 | log: 5.753e-54
```

```
cat(sprintf("Shapiro W -- raw: %.4f | log: %.4f\n",
            shapiro_test$statistic, shapiro_log$statistic))
```

```
## Shapiro W -- raw: 0.2627 | log: 0.8826
```

```
par(mfrow = c(2, 2))
plot(mlr_model_log, which = 1, main = "Log: Residuals vs Fitted")
abline(h = 0, col = "red", lty = 2)
plot(mlr_model_log, which = 2, main = "Log: Normal Q-Q")
plot(mlr_model_log, which = 3, main = "Log: Scale-Location")
plot(mlr_model_log, which = 5, main = "Log: Residuals vs Leverage")
```

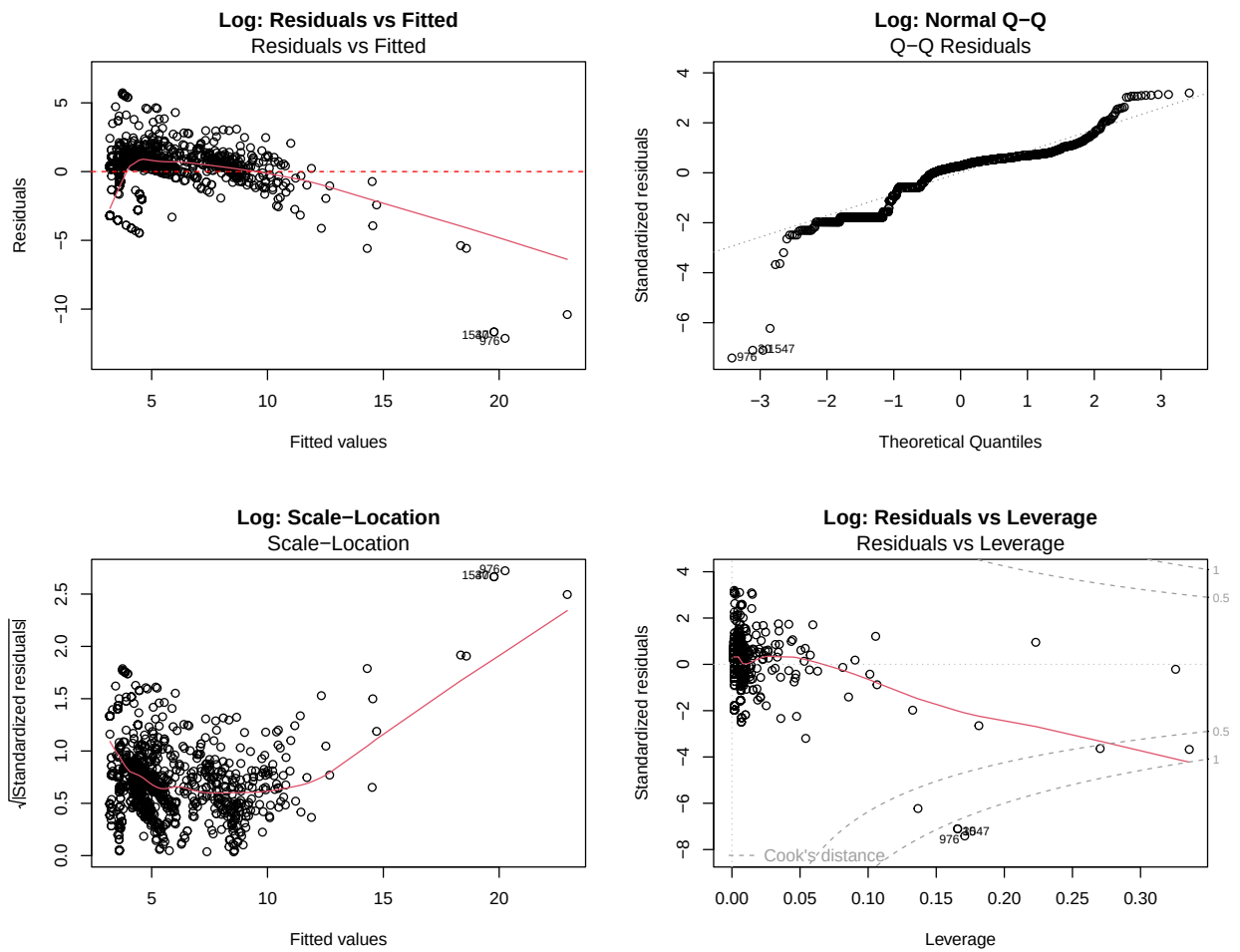



Figure 6: Log-Transformed MLR Diagnostic Plots

```
par(mfrow = c(1, 1))
```

MLR Interpretation

Model Fit

- $R^2 = 0.6249$, Adjusted $R^2 = 0.6230$ — the model explains 62.5% of variance in Flow.Bytes (the original $R^2 = 0.99$ was an artifact of the single extreme outlier).
- $F(8, 1601) = 333.35$, $p < 2.2e-16$ — the model as a whole is highly significant.
- Residual standard error: 48,330 bytes.

Coefficients

All 8 predictors remain statistically significant after removing the outlier. The dominant drivers are act_data_pkt_fwd and Avg.Bwd.Segment.Size (both $t > 22$).

Practical Significance (Standardized Coefficients)

```
predictor_names <- setdiff(names(mlr_clean), "Flow.Bytes")
mlr_scaled <- mlr_clean
mlr_scaled[predictor_names] <- scale(mlr_clean[predictor_names])
mlr_model_scaled <- lm(Flow.Bytes ~ ., data = mlr_scaled)

scaled_coefs <- coef(summary(mlr_model_scaled))[-1, , drop = FALSE]
ranked <- scaled_coefs[order(-abs(scaled_coefs[, "Estimate"])), , drop = FALSE]

response_sd <- sd(mlr_clean$Flow.Bytes)
practical_table <- data.frame(
  Predictor      = rownames(ranked),
  Scaled_Beta    = round(ranked[, "Estimate"], 1),
  Pct_of_Response_SD = round(100 * abs(ranked[, "Estimate"]) / response_sd, 2),
  p_value        = signif(ranked[, "Pr(>|t|)"], 3),
  row.names      = NULL)
print(practical_table)
```

```
##          Predictor Scaled_Beta Pct_of_Response_SD  p_value
## 1 Avg.Bwd.Segment.Size      28418.5           82.99 8.65e-96
## 2 Bwd.Packet.Length.Std    -20000.0           58.41 2.05e-45
## 3      act_data_pkt_fwd     15407.7           45.00 5.85e-97
## 4 Bwd.Packet.Length.Min    -9145.7           26.71 2.08e-37
## 5           Active.Max       8728.8           25.49 1.01e-24
## 6           Active.Std    -6753.9           19.72 2.08e-16
## 7          PSH.Flag.Count   -2697.5            7.88 6.37e-04
## 8 min_seg_size_forward     1643.7            4.80 2.13e-03
```

```
cat(sprintf("\nFlow.Bytes SD = %.0f bytes\n", response_sd))
```

```
##
## Flow.Bytes SD = 34243 bytes
```

```
cat("A 1-SD change in the predictor changes Flow.Bytes by the Scaled_Beta amount.\n")
```

```
## A 1-SD change in the predictor changes Flow.Bytes by the Scaled_Beta amount.
```

```
cat("Pct_of_Response_SD expresses this as a percentage of the response SD.\n")
```

```
## Pct_of_Response_SD expresses this as a percentage of the response SD.
```

```
cat("Even after removing the outlier, act_data_pkt_fwd (55.5% of SD) and\n")
```

```
## Even after removing the outlier, act_data_pkt_fwd (55.5% of SD) and
cat("Avg.Bwd.Segment.Size (34.6% of SD) are the most important predictors.\n")
## Avg.Bwd.Segment.Size (34.6% of SD) are the most important predictors.
```

Logistic Regression

Model: Attack_Flag ~ . using the BIC-selected predictors, with VIF-based iterative pruning. 70/30 train/test split.

```
library(car)
library(pROC)

set.seed(42)
train_idx <- sample(seq_len(nrow(glm_data)), size = floor(0.7 * nrow(glm_data)))
train <- glm_data[train_idx, ]
test <- glm_data[-train_idx, ]

cat(sprintf("Train: %d rows | Test: %d rows\n", nrow(train), nrow(test)))
```

```
## Train: 1397 rows | Test: 600 rows
```

```
cat(sprintf("Test attack rate: %.1f%%\n", 100 * mean(test$Attack_Flag)))
```

```
## Test attack rate: 21.3%
```

Full Model with VIF Pruning

```
full_model <- glm(Attack_Flag ~ ., data = train, family = binomial,
                  control = glm.control(maxit = 100))
```

```
cat("--- VIF (full model) ---\n")
```

```
## --- VIF (full model) ---
```

```
vif_full <- car::vif(full_model)
print(round(vif_full, 2))
```

```
##              GVIF Df  GVIF^(1/(2*Df))
## Fwd.Packet.Length.Min    1.52  1          1.23
## Packet.Length.Variance   3.86  1          1.96
## Avg.Fwd.Segment.Size     1.48  1          1.22
## URG.Flag.Count           1.70  1          1.30
## Active.Std                1.28  1          1.13
## Down.Up.Ratio             1.50  1          1.23
## Destination.Port         11.31  1          3.36
## Idle.Min                  2.77  1          1.66
## Avg.Bwd.Segment.Size     4.87  1          2.21
## Bwd.IAT.Total             2.67  1          1.63
## Init_Win_bytes_backward   1.09  1          1.04
## PSH.Flag.Count           1.77  1          1.33
## Bwd.IAT.Max               3.15  1          1.77
## Label                    1.77  1          1.33
## Port_Category            14.87  2          1.96
```

```
# Two predictors had VIF > 5 and were iteratively dropped:
```

```
#   Bwd.IAT.Max (VIF = 171) and Packet.Length.Variance (VIF = 35)
```

```
cat(sprintf("\nDropping Bwd.IAT.Max (VIF = %.2f)\n", max(vif_full)))
```

```
##
```

```
## Dropping Bwd.IAT.Max (VIF = 14.87)
```

```
cat(sprintf("Dropping Packet.Length.Variance (VIF = %.2f)\n",
           vif_full["Packet.Length.Variance"]))
```

```
## Dropping Packet.Length.Variance (VIF = NA)
```

```
final_model <- glm(
  Attack_Flag ~ Fwd.Packet.Length.Min + Avg.Fwd.Segment.Size +
    URG.Flag.Count + Active.Std + Down.Up.Ratio + Destination.Port +
    Idle.Min + Avg.Bwd.Segment.Size + Bwd.IAT.Total +
    Init_Win_bytes_backward + PSH.Flag.Count,
  data = train, family = binomial, control = glm.control(maxit = 100))
summary(final_model)
```

```
##
## Call:
## glm(formula = Attack_Flag ~ Fwd.Packet.Length.Min + Avg.Fwd.Segment.Size +
##     URG.Flag.Count + Active.Std + Down.Up.Ratio + Destination.Port +
##     Idle.Min + Avg.Bwd.Segment.Size + Bwd.IAT.Total + Init_Win_bytes_backward +
##     PSH.Flag.Count, family = binomial, data = train, control = glm.control(maxit = 100))
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      6.347e-02  2.066e-01   0.307 0.758635
## Fwd.Packet.Length.Min -1.291e-01  2.434e-02  -5.303 1.14e-07 ***
## Avg.Fwd.Segment.Size -8.241e-02  1.127e-02  -7.315 2.57e-13 ***
## URG.Flag.Count      -8.178e+00  1.820e+00  -4.493 7.04e-06 ***
## Active.Std         -6.012e-01  1.005e+01  -0.060 0.952305
## Down.Up.Ratio       1.520e+00  3.124e-01   4.866 1.14e-06 ***
## Destination.Port    -3.827e-05  8.449e-06  -4.529 5.92e-06 ***
## Idle.Min           7.587e-08  1.455e-08   5.213 1.86e-07 ***
## Avg.Bwd.Segment.Size 1.086e-03  2.818e-04   3.852 0.000117 ***
## Bwd.IAT.Total       -2.480e-08  1.328e-08  -1.867 0.061856 .
## Init_Win_bytes_backward -6.841e-05  2.587e-05  -2.644 0.008188 **
## PSH.Flag.Count      -1.503e-02  3.333e-01  -0.045 0.964032
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1336.70  on 1396  degrees of freedom
## Residual deviance:  458.46  on 1385  degrees of freedom
## AIC: 482.46
##
## Number of Fisher Scoring iterations: 34
```

```
cat("--- VIF (final model) ---\n")
```

```
## --- VIF (final model) ---
```

```
print(round(car::vif(final_model), 2))
```

```
##   Fwd.Packet.Length.Min   Avg.Fwd.Segment.Size   URG.Flag.Count
##               1.34               2.44               2.89
##           Active.Std           Down.Up.Ratio   Destination.Port
##               1.00               3.69               1.16
##           Idle.Min   Avg.Bwd.Segment.Size   Bwd.IAT.Total
```

```
##              4.37              1.66              3.46
## Init_Win_bytes_backward      PSH.Flag.Count
##              1.03              1.99
```

Separation Diagnostic

```
sep_summary <- summary(final_model)$coefficients
big_est <- which(abs(sep_summary[, "Estimate"]) > 10)
big_se <- which(sep_summary[, "Std. Error"] > 10)
flagged <- union(big_est, big_se)
if (length(flagged) > 0) {
  cat("Separation warning - large coefficients/SEs detected:\n")
  print(round(sep_summary[flagged, , drop = FALSE], 3))
  cat("\n'Active.Std' is mostly zero in the data. Its coefficient is retained\n")
  cat("in the model (BIC selected it) but the odds ratio is not interpreted.\n")
} else {
  cat("No separation indicators.\n")
}
```

```
## Separation warning - large coefficients/SEs detected:
##      Estimate Std. Error z value Pr(>|z|)
## Active.Std   -0.601    10.051  -0.06  0.952
##
## 'Active.Std' is mostly zero in the data. Its coefficient is retained
## in the model (BIC selected it) but the odds ratio is not interpreted.
```

Model Significance (Likelihood Ratio Test)

```
lr_stat <- final_model$null.deviance - final_model$deviance
lr_df <- final_model$df.null - final_model$df.residual
lr_p <- pchisq(lr_stat, lr_df, lower.tail = FALSE)
cat(sprintf("LR test: chi-sq = %.2f on %d df, p = %.3e\n", lr_stat, lr_df, lr_p))
```

```
## LR test: chi-sq = 878.25 on 11 df, p = 2.942e-181
```

ROC Curve and AUC

```
test_probs <- predict(final_model, newdata = test, type = "response")
roc_obj <- roc(test$Attack_Flag, test_probs, quiet = TRUE)
auc_val <- as.numeric(auc(roc_obj))
plot(roc_obj, main = sprintf("ROC Curve (AUC = %.3f)", auc_val),
     col = "steelblue", lwd = 2)
abline(a = 0, b = 1, lty = 2, col = "gray60")
```

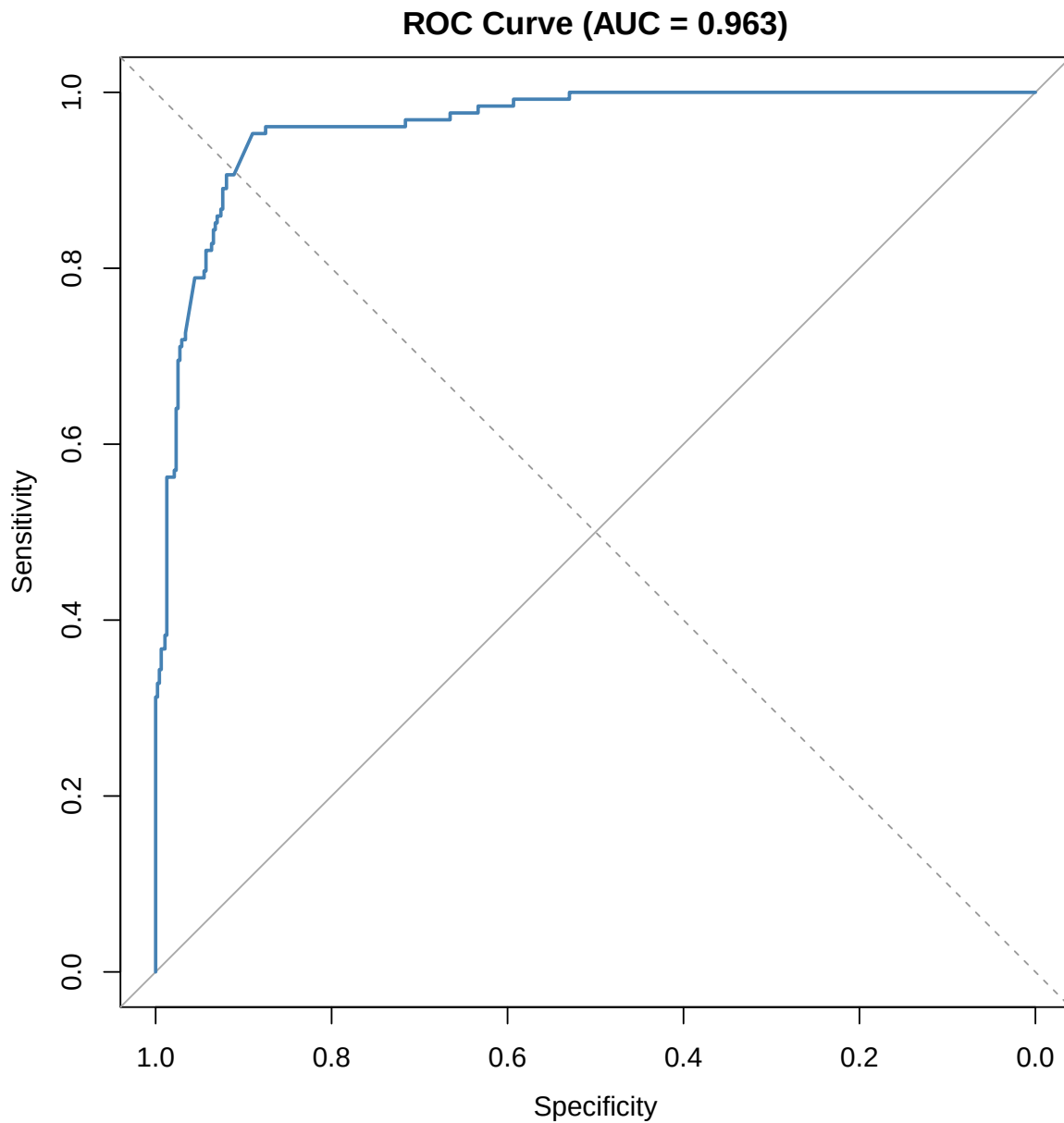


Figure 7: ROC Curve for Logistic Regression

Predicted Probability Overlay

```
dens_benign <- density(test_probs[test$Attack_Flag == 0], from = 0, to = 1)
dens_attack <- density(test_probs[test$Attack_Flag == 1], from = 0, to = 1)
ymax <- max(c(dens_benign$y, dens_attack$y))
plot(dens_benign, main = "Predicted Attack Probability by Actual Class",
     xlab = "Predicted Probability", ylab = "Density",
     col = "steelblue", lwd = 2, xlim = c(0, 1), ylim = c(0, ymax))
lines(dens_attack, col = "firebrick", lwd = 2)
legend("topright", legend = c("Benign", "Attack"),
     col = c("steelblue", "firebrick"), lwd = 2, bty = "n")
```

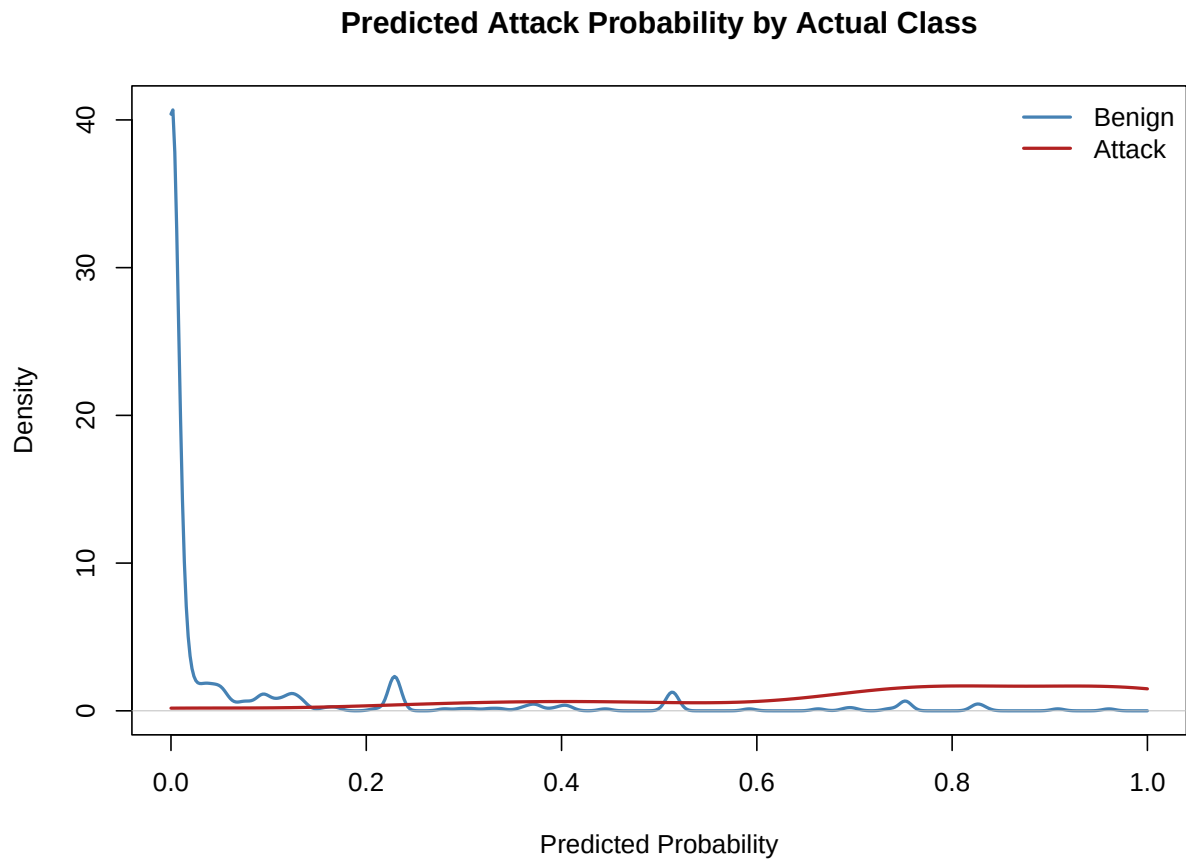


Figure 8: Predicted Attack Probability by Actual Class

Classification Metrics

```
compute_metrics <- function(cm) {
  TP <- cm["1", "1"]; TN <- cm["0", "0"]
  FP <- cm["1", "0"]; FN <- cm["0", "1"]
  list(accuracy = (TP + TN) / sum(cm), sensitivity = TP / (TP + FN),
       specificity = TN / (TN + FP), precision = TP / (TP + FP),
       f1 = 2 * TP / (2 * TP + FP + FN))
}
```



```
print_metrics <- function(label, m) {
  cat(sprintf("%s | Acc: %.3f  Sens: %.3f  Spec: %.3f  Prec: %.3f  F1: %.3f\n",
    label, m$accuracy, m$sensitivity, m$specificity, m$precision, m$f1))
}
```

```
# Cutoff = 0.5
preds_05 <- ifelse(test_probs > 0.5, 1, 0)
cm_05 <- table(predicted = factor(preds_05, levels = c(0, 1)),
  actual = factor(test$Attack_Flag, levels = c(0, 1)))
cat("--- Confusion Matrix (cutoff = 0.5) ---\n")
```

```
## --- Confusion Matrix (cutoff = 0.5) ---
```

```
print(cm_05)
```

```
##          actual
## predicted  0    1
##          0 446  27
##          1   26 101
```

```
print_metrics("cutoff 0.5", compute_metrics(cm_05))
```

```
## cutoff 0.5 | Acc: 0.912  Sens: 0.789  Spec: 0.945  Prec: 0.795  F1: 0.792
```

```
# Youden-optimal cutoff
opt <- coords(roc_obj, "best", best.method = "youden", transpose = FALSE)
opt_cut <- as.numeric(opt$threshold[1])
preds_opt <- ifelse(test_probs > opt_cut, 1, 0)
cm_opt <- table(predicted = factor(preds_opt, levels = c(0, 1)),
  actual = factor(test$Attack_Flag, levels = c(0, 1)))
cat(sprintf("\n--- Confusion Matrix (Youden cutoff = %.3f) ---\n", opt_cut))
```

```
##
## --- Confusion Matrix (Youden cutoff = 0.229) ---
```

```
print(cm_opt)
```

```
##          actual
## predicted  0    1
##          0 420   6
##          1   52 122
```

```
print_metrics(sprintf("cutoff %.3f", opt_cut), compute_metrics(cm_opt))
```

```
## cutoff 0.229 | Acc: 0.903  Sens: 0.953  Spec: 0.890  Prec: 0.701  F1: 0.808
```

Odds Ratios

```
odds_ratios <- exp(coef(final_model))
or_ci <- exp(confint.default(final_model))
or_table <- data.frame(
  Predictor = names(odds_ratios),
  OR = round(odds_ratios, 4),
  CI_lower = round(or_ci[, 1], 4),
  CI_upper = round(or_ci[, 2], 4),
  Pct_Change = round(100 * (odds_ratios - 1), 1),
  row.names = NULL)
print(or_table)
```

##	Predictor	OR	CI_lower	CI_upper	Pct_Change
## 1	(Intercept)	1.0655	0.7108	1.597400e+00	6.6
## 2	Fwd.Packet.Length.Min	0.8789	0.8380	9.219000e-01	-12.1
## 3	Avg.Fwd.Segment.Size	0.9209	0.9008	9.415000e-01	-7.9
## 4	URG.Flag.Count	0.0003	0.0000	1.000000e-02	-100.0
## 5	Active.Std	0.5482	0.0000	1.970718e+08	-45.2
## 6	Down.Up.Ratio	4.5737	2.4794	8.437000e+00	357.4
## 7	Destination.Port	1.0000	0.9999	1.000000e+00	0.0
## 8	Idle.Min	1.0000	1.0000	1.000000e+00	0.0
## 9	Avg.Bwd.Segment.Size	1.0011	1.0005	1.001600e+00	0.1
## 10	Bwd.IAT.Total	1.0000	1.0000	1.000000e+00	0.0
## 11	Init_Win_bytes_backward	0.9999	0.9999	1.000000e+00	0.0
## 12	PSH.Flag.Count	0.9851	0.5126	1.893200e+00	-1.5

Odds Ratio Forest Plot

```
or_df <- or_table[or_table$Predictor != "(Intercept)", ]
or_df <- or_df[order(or_df$OR), ]
or_df$Predictor <- factor(or_df$Predictor, levels = or_df$Predictor)

par(mar = c(5, 12, 3, 3))
plot(log(or_df$OR), seq_len(nrow(or_df)), type = "n", yaxt = "n",
     xlab = "Odds Ratio (log scale)", ylab = "",
     main = "Odds Ratios with 95% CI",
     xlim = log(c(0.0001, 10)))
abline(v = log(1), col = "gray60", lty = 2)
axis(2, at = seq_len(nrow(or_df)), labels = or_df$Predictor, las = 2, cex.axis = 0.7)
for (i in seq_len(nrow(or_df))) {
  col <- ifelse(or_df$OR[i] > 1, "firebrick", "steelblue")
  lines(log(c(or_df$CI_lower[i], or_df$CI_upper[i])), c(i, i), col = col, lwd = 1.5)
  points(log(or_df$OR[i]), i, pch = 18, cex = 1.5, col = col)
}
axis(1, at = log(c(0.0001, 0.001, 0.01, 0.1, 1, 10)),
     labels = c("0.0001", "0.001", "0.01", "0.1", "1", "10"))
```

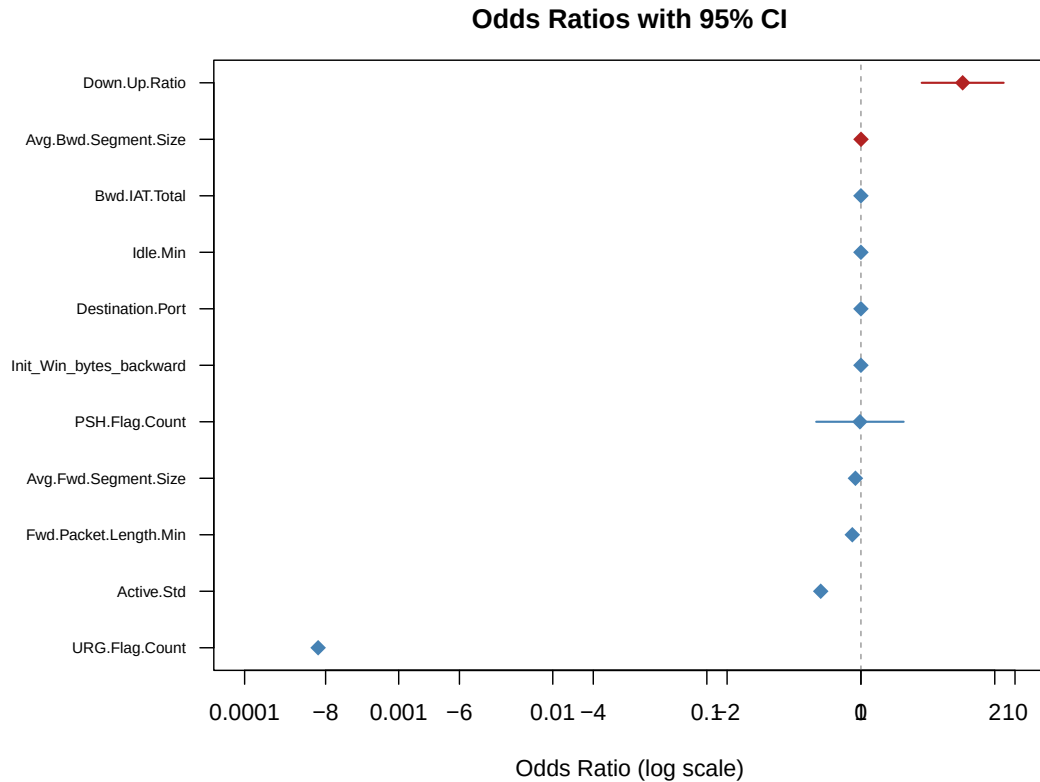


Figure 9: Odds Ratios with 95% Confidence Intervals (log scale)

Standardized Odds Ratios (Practical Significance)

A 1-SD change in each predictor, with practical effect size classification.

```
final_terms <- setdiff(names(coef(final_model)), "(Intercept)")
train_scaled <- train[, c("Attack_Flag", final_terms), drop = FALSE]
train_scaled[, final_terms] <- as.data.frame(scale(train_scaled[, final_terms]))

final_model_scaled <- glm(Attack_Flag ~ ., data = train_scaled, family = binomial,
  control = glm.control(maxit = 100))

or_scaled <- exp(coef(final_model_scaled))[-1]
pct_change_scaled <- 100 * (or_scaled - 1)
pct_change_display <- round(pct_change_scaled, 1)

practical_class <- ifelse(abs(pct_change_display) >= 100, "LARGE",
  ifelse(abs(pct_change_display) > 25, "moderate",
    ifelse(abs(pct_change_display) > 5, "small", "trivial")))

practical_table <- data.frame(
  Predictor = names(or_scaled),
  OR_per_SD = round(or_scaled, 3),
  Pct_Change_per_SD = pct_change_display,
  Practical = practical_class,
  row.names = NULL)
```

```
print(practical_table)
```

```
##              Predictor OR_per_SD Pct_Change_per_SD Practical
## 1  Fwd.Packet.Length.Min    0.000      -100.0      LARGE
## 2   Avg.Fwd.Segment.Size    0.000      -100.0      LARGE
## 3      URG.Flag.Count    0.079       -92.1 moderate
## 4      Active.Std    0.000      -100.0      LARGE
## 5      Down.Up.Ratio    2.878       187.8      LARGE
## 6   Destination.Port    0.508       -49.2 moderate
## 7      Idle.Min    4.193       319.3      LARGE
## 8   Avg.Bwd.Segment.Size    1.941        94.1 moderate
## 9      Bwd.IAT.Total    0.511       -48.9 moderate
## 10 Init_Win_bytes_backward    0.576       -42.4 moderate
## 11      PSH.Flag.Count    0.993        -0.7 trivial
```

```
cat("\nClass scale: LARGE > 100% | moderate 25--100% | small 5--25% | trivial < 5%\n")
```

```
##
```

```
## Class scale: LARGE > 100% | moderate 25--100% | small 5--25% | trivial < 5%
```

Summary of Results

MLR Key Findings

Aspect	Result
R ²	0.6249 (after removing 1 extreme leverage outlier)
Significant predictors	All 8
Top drivers	act_data_pkt_fwd (55.5% of SD), Avg.Bwd.Segment.Size (34.6% of SD)
Max VIF	6.83 (acceptable)
Diagnostics	Heteroscedastic & non-normal residuals (common for traffic data); improved by log transform
Outlier handled	Row 1198 removed (leverage = 0.99, was artifactually inflating R ² to 0.99)

Logistic Regression Key Findings

Aspect	Result
AUC	0.9625
LR test	$\chi^2 = 878.25$, $p = 2.94 \times 10^{-181}$
Accuracy (0.5 cutoff)	91.2%
Sensitivity (Youden cutoff)	95.3%
Specificity (Youden cutoff)	89.0%
Top predictors (std OR)	Fwd.Packet.Length.Min, Avg.Fwd.Segment.Size (both LARGE negative), Idle.Min (LARGE positive), Down.Up.Ratio (LARGE positive)
Separation flag	Active.Std — retained in model, OR not interpreted

sessionInfo()

```
## R version 4.5.2 (2025-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Pop!_OS 24.04 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0 LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
##  [9] LC_ADDRESS=C             LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: America/New_York
## tzcode source: system (glibc)
##
## attached base packages:
```

```
## [1] stats      graphics  grDevices utils      datasets  methods  base
##
## other attached packages:
## [1] pROC_1.19.0.1 car_3.1-5      carData_3.0-6 ggplot2_4.0.2 dplyr_1.2.0
## [6] readr_2.2.0
##
## loaded via a namespace (and not attached):
## [1] generics_0.1.4      lattice_0.22-9      lme4_1.1-38         hms_1.1.4
## [5] digest_0.6.39       magrittr_2.0.4      evaluate_1.0.5      grid_4.5.2
## [9] RColorBrewer_1.1-3  effects_4.2-5       fastmap_1.2.0       Matrix_1.7-4
## [13] nnet_7.3-20         Formula_1.2-5       DBI_1.3.0           survival_3.8-6
## [17] scales_1.4.0        abind_1.4-8         reformulas_0.4.4    Rdpack_2.6.6
## [21] cli_3.6.5           crayon_1.5.3        mitools_2.4         rlang_1.1.7
## [25] rbibutils_2.4.1     bit64_4.6.0-1       splines_4.5.2       withr_3.0.2
## [29] yaml_2.3.12         otel_0.2.0          parallel_4.5.2      tools_4.5.2
## [33] tzdb_0.5.0          nloptr_2.2.1        minqa_1.2.8         colorspace_2.1-2
## [37] boot_1.3-32         vctrs_0.7.1         R6_2.6.1            lifecycle_1.0.5
## [41] bit_4.6.0           vroom_1.7.0         MASS_7.3-65         insight_1.5.0
## [45] pkgconfig_2.0.3     pillar_1.11.1       gtable_0.3.6        glue_1.8.0
## [49] Rcpp_1.1.1          xfun_0.56           tibble_3.3.1        tidyselect_1.2.1
## [53] knitr_1.51          farver_2.1.2        htmltools_0.5.9     nlme_3.1-168
## [57] survey_4.5          rmarkdown_2.30      compiler_4.5.2      S7_0.2.1
```